🏙️ Cell City Builders

Name: Class: Date:

Activity 1 · Hands-on · 🧪 In the lab · ⏱ 30 minutes

🧅 Onion Peel Cells

You will learn: Plant cells have a cell wall, a nucleus and a large vacuole — you can see and measure them.

YOU NEED

- An onion
- Iodine solution (or a drop of tea decoction)
- Glass slide and cover slip (or clear sheet)
- Forceps or a needle
- Blotting paper
- Phone microscope

⚠️ **Safety:** Iodine stains skin and clothes; use a dropper.

STEPS

1. Peel the thin transparent layer from the inside of an onion scale.
2. Place a small flat piece on the slide with a drop of water.
3. Add a drop of iodine and wait one minute.
4. Lower the cover slip gently to avoid air bubbles; blot extra liquid.
5. Observe at low zoom, then at high zoom. Use 📏 Set scale to measure one cell.
6. Draw 3–4 cells and label the cell wall, nucleus and cytoplasm.

OBSERVATION TABLE

Structure	Seen? (yes/no)	Description

QUESTIONS

1. Why are onion cells arranged like bricks?

2. Why did we use iodine?

3. Which cell organelle was not visible? Why?

WHAT IS HAPPENING? — THE SCIENCE

The brick-like cells are joined edge to edge. The rigid cellulose cell wall gives shape; the stained dark dot is the nucleus; the large central vacuole pushes the cytoplasm to the edge. These features are typical of plant cells.

Key terms: cell wall · nucleus · vacuole · cytoplasm

Activity 2 · Hands-on · 🏠 At home · ⏱ 1–2 hours

📦 Build a 3D Cell Model

> **You will learn:** Each organelle has a job — your model should show what each one does.

YOU NEED

- A box or clear plastic container
- Coloured clay, beads, buttons, rubber bands, string, bubble wrap
- Paper labels and toothpicks

> ⚠️ **Safety:** Use scissors and glue carefully.

STEPS

1. Choose plant cell or animal cell.
2. Plan which material will stand for each organelle and why (e.g., a folded ribbon for cristae).
3. Build the nucleus with a nucleolus, then ER, Golgi, mitochondria, ribosomes and lysosomes (and chloroplasts plus a vacuole for a plant cell).
4. Add a label flag to each organelle with its function.
5. Present your model and explain the "city job" of each part.

OBSERVATION TABLE

Organelle	Material used	Why I chose it	Function

QUESTIONS

1. Which organelles have their own DNA?

2. Which organelle is the "post office" and why?

3. Name two differences between your plant and animal cell models.

WHAT IS HAPPENING? — THE SCIENCE

Linking a material to a function helps memory: folded ribbon = cristae (more surface for ATP making), stacked coins = grana, bubble wrap = ER with ribosomes. Membrane-bound organelles divide work inside eukaryotic cells.

Key terms: **mitochondria** **chloroplast** **Golgi apparatus** **endoplasmic reticulum** **lysosome**

Digital activity

💻 Organelle Job Fair

> Tap a job, then tap the organelle that does it.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu1-ch8/2**

MITOCHONDRION · CHLOROPLAST · GOLGI APPARATUS · LYSOSOME · RIBOSOME · NUCLEOLUS

Mitochondrion	Chloroplast	Golgi apparatus	Lysosome	Ribosome	Nucleolus

Makes ATP by aerobic respiration · Has cristae · Traps light energy · Has grana of thylakoids · Packs and sends out materials · Forms glycoproteins · "Suicide bag" of hydrolytic enzymes · Digests worn-out organelles · Site of protein synthesis · 70S or 80S particle · Makes rRNA · Dense body inside the nucleus

> **You will learn:** Matching structure to function is the key to this chapter: folds and stacks increase surface area; enzymes are kept in membrane bags.

THINK & WRITE

1. Which two organelles are called semi-autonomous?

2. Why are lysosomal enzymes kept inside a membrane?